MWALILI BRIAN MGHANGA

Energy Technology & Engineering Student

brianmwalili@gmail.com | +254 741 547 925 | Kenya | linkedin.com/in/brian-mwalili

PROFESSIONAL SUMMARY

Final-year BSc Energy Technology and Engineering student at Kenyatta University, committed to advancing the clean energy transition in Africa and beyond. Trained across solar, wind, hydro, bio-energy, energy storage and energy policy, with hands-on experience in power distribution and substation operations (Kenya Power attachment) and in designing a solar PV and Arduino-based system that cuts wasted lighting energy. AutoCAD and Revit certified. Keen to help develop practical, affordable renewable energy solutions for communities.

EDUCATION

- **BSc Energy Technology and Engineering**, Kenyatta University (Sept 2022 – 2027)
- **Certificate in Revit**, Bell Institute of Technology (Jun 2023 – Sept 2023)
- **Certificate in AutoCAD (Computer Aided Drawing)**, Bell Institute of Technology (Apr 2023 – Jun 2023)
- **Kenya Certificate of Secondary Education**, Oloolaiser Boys High School (Jan 2018 – Apr 2022)

RELEVANT COURSEWORK

- **Renewable energy:** Photovoltaic Technology, Solar Power Plants, Solar Thermal Energy, Wind Energy, Hydropower, Geothermal Energy, Bio-Energy, Biogas Plant Systems
- **Power and storage:** Electric Power Systems and Applications, Energy Storage Systems and Distribution, Engineering Electronics
- **Energy management and policy:** Energy Management, Energy Planning, Policy and Trading, Environmental Impact Assessment and Audit
- **Engineering and professional:** Systems Methods and Control Engineering, Metrology and Instrumentation, Project Management, Entrepreneurship, Occupational Health and Safety, Project I and II

WORK EXPERIENCE

Industrial Attachment – Distribution & Substations, Kenya Power and Lighting Company (KPLC) **May 2025 – Aug 2025**

- Shadowed technicians on daily power distribution and substation operations.
- Assisted with substation inspections and fault response activities.
- Documented inspection findings and maintenance records.
- Followed and reinforced electrical safety procedures in the field.

FINAL YEAR PROJECT

Arduino-Based Solar PV System to Minimise Energy Consumption, Kenyatta University **2026**

- Designed and built a small-scale solar PV lighting prototype: 3.5 W panel, 12 V 7 Ah battery and PWM charge controller powering an Arduino-controlled LED load.
- Integrated two LDR light sensors (indoor and outdoor) and a PIR motion sensor so lighting switches on only when a room is both dark and occupied; live readings shown on a 16x2 LCD.
- Programmed the controller in Arduino C++, calibrated light thresholds across daylight, overcast, evening and night conditions, and tested the system over a one-week period.

TECHNICAL SKILLS

- **Design software:** AutoCAD (certified, practical drafting and design experience); Revit (certified)
- **Renewable energy:** Solar PV system sizing and assembly, battery storage, charge controllers
- **Electronics and control:** Arduino programming for sensor-based control, sensor interfacing (LDR, PIR), LCD displays, circuit prototyping
- **Power systems:** Distribution and substation operations, inspection and fault response, electrical safety
- **Creative technology:** Augmented reality art using Lens Studio; AR art videos published on TikTok (@ar.ke15)
- **Software:** Microsoft Office (Word, Excel, PowerPoint)

VOLUNTEER WORK

Volunteer, Career Mothers for Exclusive Breastfeeding Organization (CAMFEB) **Aug 2024**

- Helped organise the breastfeeding run held at Waterfront, Karen.

LANGUAGES

English – Fluent | Swahili – Fluent